GRADE 2 · DEBUGGING & READING CODE

The Wrong Number

A move has the wrong count — fix it

Name: .... Date: ....

★ **I can find a move with the wrong number and fix it so both scripts reach the goal.**

The goal: both reach 4

Bit (green) and Pixel (purple) both start on the green flag. They should BOTH reach number 4. One script is correct, but the other has a move with the WRONG NUMBER, so it lands short. Read both, find the wrong number, and fix it.

The goal: both reach 4

0	1	2	3	4	5
Pixel	Bit			★ goal	

Bit's script (correct)

⚑ when green flag clicked

forward 3

Pixel's script (has a bug)

⚑ when green flag clicked

forward 3

1 Run both

Where does each land? Bit (home 1, forward 3): ____ Pixel (home 0, forward 3): ____. Which one misses the goal 4?

____

____

2 Find the wrong number

Pixel starts at 0 and needs to reach 4. Its move says 'forward 3'. What number SHOULD the move be?

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3 Fix it

Write Pixel's fixed move: forward ____. After the fix, where does Pixel land? Do both reach the goal now?

Big idea

Sometimes a script has the right blocks but the wrong NUMBER inside one of them. Find which move falls short of the goal, count how far it needs to go, and change the number. Here Pixel needed forward 4, not forward 3.

TEACHER GUIDE · GRADE 2 · DEBUGGING & READING CODE

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Quick facilitation notes & answer key

What this worksheet teaches

Students fix a wrong-number bug in one of two sets of moves. The goal is for both to reach 4; Bit is right, but Pixel's "forward 3" falls one short. Students find the buggy set and work out the correct number. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel) so children can trace both sets of moves and see which one misses the goal.

How to lead it (about 20 minutes)

1. Show them (you do). Pixel needs to reach 4 from 0 — count the steps to find the right number.
2. Do one together. Exercise 1 — Bit reaches 4, Pixel lands on 3 (the buggy one).
3. Let them try. Students work out the correct number and re-run (Ex 2–3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 1, forward 3 → 4 (reaches goal). Pixel: 0, forward 3 → 3 (misses; Pixel is buggy).

Exercise 2 — Pixel starts at 0 and needs 4, so the move should be forward 4.

Exercise 3 — fixed move forward 4 → Pixel lands on 4. Both reach 4. Only the number changes; the direction and Bit's set stay the same.

If students get stuck — and ways to adjust

Watch for: changing the direction instead of the number. Here only the number is wrong.

Make it easier: count the squares from Pixel's home to the goal.

Make it harder: ask what number would overshoot the goal.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, find and fix the wrong one, and explain the fix.

You'll know it worked when

The child fixes Pixel's number so both reach the goal.